The growth law of the last fall degree in the support width: what is proved, what is fitted

2026-11-12 · `research/prime_ic_leads_20261110/growth_law/` · derivation; data from `devcheck/` rows (development checks, `[pcc:unverified]`); extends `prime_fall_degree_20261027/DERIVED.md` D1–D8

1. Answer

Four results are proved; the growth itself is fitted, not derived. **Theorem 1:** for a fixed system, d_F is non-decreasing in the tower height t (reduction modulo $z^{2^t}-1$ is a filtered ring map sending variables to variables or 1, and closures map into closures). **Theorem 2:** a degree-c closure uses the top relation $y_{t-1}^2=1$ only through the top $c-1$ variables (a carry leaves the tower only if bits $j..t-1$ are all set). **Theorem 3:** d_F and ffd have generic values determined by the support; the normal form of S_3 fills the width-2 box exactly, so "null equals tower" means S_3 sits in that box's generic stratum. **Theorem 4:** ffd $=3,4,5,5$ for widths $W=1,2,3,4$ (a common variable in the top form forces a fall one degree above it), equal to every measured row, and the reason $W=3$ and $W=4$ coincide.

2. The fitted law (Conjecture G, $m=2$)

$$d_F(W,t)=\mathrm{ffd}(W)+\max\left(0,\left\lfloor\frac{t-s_W}{4}\right\rfloor\right),\quad s_1=0,s_2=2,s_3=s_4=5$$

It matches every completed cell ($W=1$: $t=2..11$; $W=2$: $t=2..12$, which is C-step; $W=3,4$: $t=2..10$, censored cells as lower bounds). Period four tower levels at every width, i.e. slope $1/8$ per unknown bit. **Falsifying predictions (not yet run):** $W=1$ reads 6 at $t=12$; $W=3$ reads 6 at $t=10..12$ and 7 at $t=13$; $W=2$ reads 7 at $t=14$.

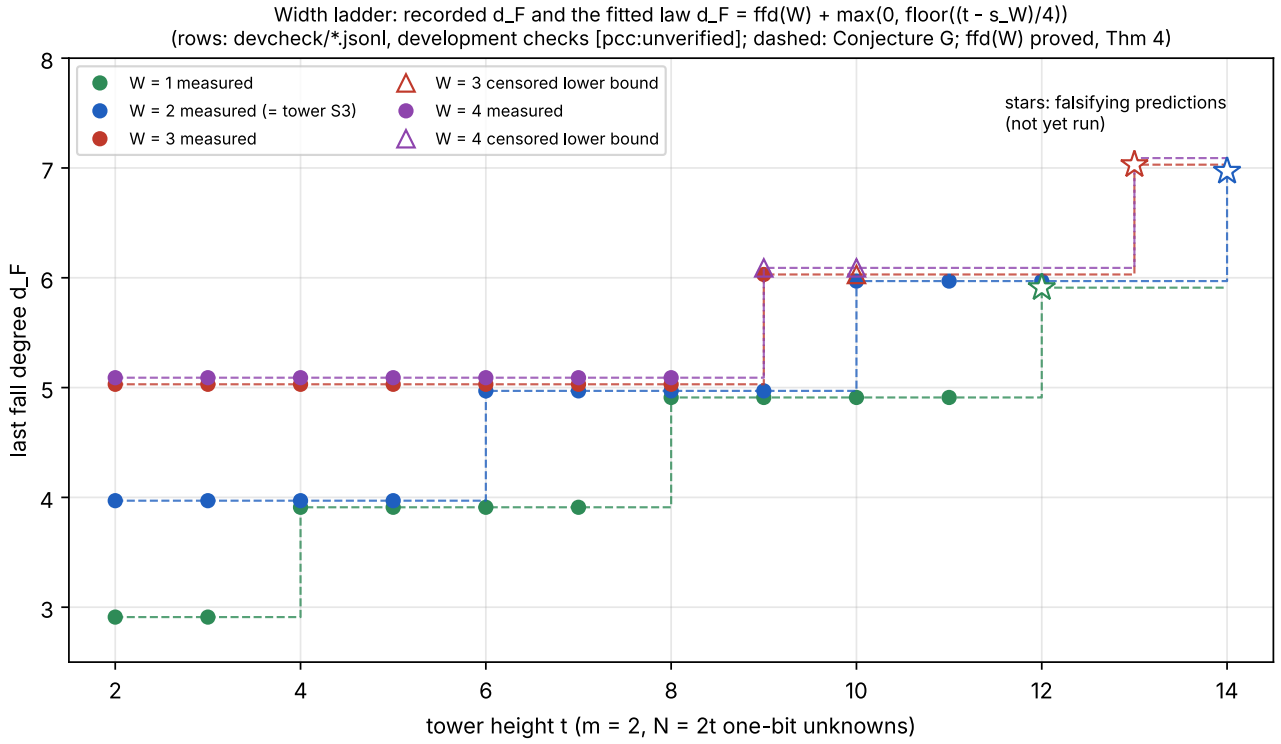


Figure 1: Recorded last fall degrees of the width ladder ($m=2$, Kummer at P_1, two random targets per cell; triangles: censored lower bounds) and the fitted law (dashed). Stars: the falsifying predictions.

3. Statements and proofs (summary; full text in README)

Theorem 1. $\varphi: R_{t+1}\to R_t$, $z^e\mapsto z^{e \bmod 2^t}$, is a ring homomorphism with $\mathrm{pop}(e \bmod 2^t)\le\mathrm{pop}(e)$; it maps $y_j\mapsto y_j$ $(j<t)$, $y_t\mapsto 1$, and $f^{(t+1)}\mapsto f^{(t)}$. The set of $u\in V_c\big(f^{(t+1)}\big)$ with $\varphi(u)\in V_c\big(f^{(t)}\big)$ satisfies the closure property, so it is all of $V_c\big(f^{(t+1)}\big)$; hence $1\in V_c\big(f^{(t+1)}\big)\Rightarrow 1\in V_c\big(f^{(t)}\big)$.

Theorem 2. $e+2^j\ge 2^t$ iff bits $j..t-1$ of e are set, so $\mathrm{pop}(e)\ge t-j$; the closure multiplies only elements of degree $<c$, so $j\ge t-c+1$. Over the twisted algebra $F_p(\zeta)[z]/(z^K-\zeta)$ the closure is identical outside this window, and a curve point with $z^K=\zeta$ is a true solution; a growing lower bound needs a functional that corrects such points inside the top window. Open.

Theorem 3. The closure is a finite tree of rank decisions on matrices polynomial in the coefficients; the branches are constructible and partition affine space, so one is dense open. $S_3(gz_1, gz_2, x_R)$ has the nine coefficients g^4, $-2x_R g^3$ (×2), $x_R^2 g^2$ (×2), $-2(x_R^2 + a)g^2$, $-(2ax_R + 4b)g$ (×2), $a^2 - 4bx_R$: generically all non-zero, so its support is the full $\{0,1,2\}^2$ box. No probability bound is claimed.

Theorem 4. If every monomial of the top form contains one variable y, then yf falls (because $y^2 = 0$ in gr R) and is not in $V_D = \langle f \rangle$ because f is a unit, so ffd $= D + 1$. $W = 1$: top $y_{10}y_{20}$, $D = 2$. $W = 3, 4$: the only popcount-2 exponent in $[0, W]$ is 3, top $y_{10}y_{11}y_{20}y_{21}$, $D = 4$. $W = 2$: D3′s bilinear argument.

4. Open, and formalization

Open: a lower bound growing with t (Theorem 2 says where it must live), the period 4, the offsets s_W, isogeny towers. Lean: not done; the container has no Lean toolchain, and the Mathlib cache exceeds the disk allowance. Theorem 1 and the Lemma of Theorem 4 are short Lean targets for a session with the `crypto-lean` gate. Knowledge base unavailable in this session.